

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR CONTROLLED COPOLYMERIZATION, SOLVENT POLYMERIZATION, AND NBR-CONTAINING FORMULATIONS

1. ABSTRACT

This report surveys a five-patent landscape relevant to low-acrylonitrile nitrile butadiene rubber (NBR) and closely related NBR-containing materials. The target is defined as NBR grades with reduced acrylonitrile content, typically below about 20 wt% ACN, where monomer ratio control, feed strategy, and polymerization conditions are used to tune composition, molecular weight, and processability. The selected patents span direct NBR synthesis in organic solvent, emulsion-based grafting and SAN/ABS-type systems that incorporate butadiene-acrylonitrile rubber domains, and formulation patents that use carboxyl-terminated butadiene-acrylonitrile rubber as a toughener in epoxy composites. Although not all patents are dedicated to low-ACN NBR, they collectively disclose the process levers most relevant to controlling acrylonitrile incorporation and downstream performance.

US9023914B2 is the most directly relevant synthesis disclosure: it describes free-radical solution polymerization of butadiene and acrylonitrile in organic solvent with modifier substances, using explicit monomer ratios, initiator loading, temperature, pressure, reaction time, and precipitation isolation. The examples show a 37/63 phm ACN/butadiene feed, 100 °C polymerization, 3 bar butadiene addition, and 9 h reaction time, with variants adjusting modifier and solvent to tune molecular weight and conversion. US9624333B2 discloses emulsion graft copolymerization of styrene/acrylonitrile onto agglomerated butadiene rubber substrates, with explicit SAN feed composition, temperatures, initiator addition, and precipitation conditions; it is relevant as a process analogue for acrylonitrile incorporation control in rubber-rich systems. US9909003B2 and US10947337B2 are formulation patents that explicitly use butadiene acrylonitrile rubber or CTBN-type tougheners in epoxy systems, showing how NBR chemistry is leveraged in composite and adhesive applications. US11813892B2 is less directly related to NBR, but it provides controlled radical polymerization and monomer-functionalization concepts that are useful as broader process references for architecture control.

Overall, the evidence supports solution or emulsion copolymerization with careful monomer feed control, moderate-to-high temperatures, and chain-transfer/modifier management as the main

routes to low-ACN NBR. The most actionable synthesis parameters are those in US9023914B2, while the other patents mainly inform downstream use and formulation compatibility.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9023914B2
- Patent Title: Nitrile rubbers and production thereof in organic solvents
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses free-radical polymerization of acrylonitrile and 1,3-butadiene in an organic solvent in the presence of specific modifier compounds, followed by precipitation isolation of the nitrile rubber. It emphasizes solution polymerization rather than conventional emulsion processing and allows tuning of molecular weight and polydispersity through modifier and solvent selection. - Acrylonitrile feed: 43 ml (653.3 mmol, corresponding to 37 phm) — Example 1 - 1,3-Butadiene feed: 90 ml (1077 mmol, corresponding to 63 phm) — Example 1 - Solvent: 50 ml (51 phm) dimethylacetamide — Example 1 - Initiator: Vazo® 88 284.9 mg (1.17 mmol, corresponding to 0.31 phm) — Example 1 - Modifier: DoPAT 204 mg (0.583 mmol, corresponding to 0.22 phm) — Example 1 - Butadiene addition pressure: 3 bar — Example 1 - Polymerization temperature: 100° C. — Example 1 - Polymerization time: 9 hours — Example 1 - Degassing time: 10 minutes with argon — Example 1 - Precipitation solvent: ethanol containing pyrocatechol/hydroquinone stabilizer — Example 1 - Final conversion: 39% — Example 6 - Reaction time: 6 hours — Example 6 - Modifier amount: 2.22 phm — Example 7 - Final conversion: 48% — Example 7 - Low molecular mass NBR Mn: 12 000 g/mol — Example 8 - Styrene feed: 2.79 g (26.8 mmol, corresponding to 50 phm) — Example 8 - Reaction temperature: 100° C. — Example 8 - Reaction time: 13 hours — Example 8 - Final conversion: 9.2% — Example 8

Relevant Experimental Evidence - Example 1: Demonstrated solution polymerization of NBR in dimethylacetamide using 37 phm acrylonitrile and 63 phm butadiene at 100° C. and 3 bar. - Example 6: Showed that the same solution process could incorporate hydroxyethyl acrylate as a termonomer, with 39% final conversion after 6 hours. - Example 7: Demonstrated deliberate lowering of molecular mass by increasing modifier amount to 2.22 phm, reaching 48% conversion after 9 h. - Example 8: Used the low-Mn NBR from Example 7 as a RAFT macromer for

block-styrene extension at 100° C. - Example 2: Not enough structured detail was provided in the extraction, but the patent indicates similar solution-polymerization variants with changed modifier/solvent conditions.

Technical Relevance This is the most directly relevant patent for low-acrylonitrile NBR because it explicitly discloses acrylonitrile/butadiene feed ratios and a controllable solution polymerization route. The examples show how modifier loading, solvent choice, temperature, and time can be used to tune composition and molecular weight in NBR synthesis.

Patent 2

- Patent Number: US9624333B2
- Patent Title: Thermoplastic moulding materials based on acrylonitrile, styrene and butadiene
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses preparation of SAN-grafted rubber materials by agglomerating a butadiene rubber graft substrate and then graft polymerizing styrene/acrylonitrile onto the agglomerated latex. The process is carried out in aqueous emulsion with persulfate initiation, controlled temperature ramping, and subsequent precipitation and washing. - Potassium persulfate: 0.074 part by weight dissolved in 3.13 parts by weight demineralized water — General procedure - Styrene feed: 32.8 parts by weight — General procedure - Acrylonitrile feed: 8.2 parts by weight — General procedure - Monomer addition time: 2 hours and 44 minutes — General procedure - Initial temperature: 68° C. — General procedure - Polymerization temperature: 80° C. — General procedure - Post-addition polymerization time: 80 minutes — General procedure - Stabilizer dispersion: 0.37 part by weight, solids content 60 wt % — General procedure - Precipitation agent: 313 parts by weight magnesium sulfate solution, solids content 1.0 wt % — General procedure - Precipitation temperature: 80° C. to 95° C. — General procedure - Washing: 2 times with 550 parts by weight demineralized water — General procedure - Drying temperature: 70° C. — General procedure - Residual moisture: less than 1 wt % — General procedure - SAN ratio of styrene to acrylonitrile: 75:25 — Abstract/description - Viscosity number: 64 ml/g — SAN polymer description - Melt flow rate: 64 ml/10 min at 220° C. with 10 kg load — SAN polymer description - Extrusion temperature: 200° C. to 250° C. — Thermoplastic molding compounds - Extruder speed: 700 rpm — Thermoplastic molding compounds - Batch size: 4 kg — Thermoplastic molding compounds

Relevant Experimental Evidence - Example 1: Demonstrated graft copolymer B preparation by SAN grafting onto agglomerated rubber latex with controlled temperature and persulfate initiation. - Example 2: Showed that the resulting ABS-type molding compounds were processed in a twin-screw extruder at 200° C. to 250° C. and 700 rpm. - Example 3: Compared

agglomeration/graft combinations and showed property differences in impact strength and flow. - Comparative example x: Provided agglomeration yield and particle-size metrics for different graft substrate/agglomerating copolymer combinations. - General procedure: Demonstrated precipitation of the graft latex with magnesium sulfate solution at elevated temperature.

Technical Relevance This patent is relevant because it discloses acrylonitrile incorporation into rubber-rich graft systems and shows how feed composition and emulsion processing are controlled. While not a direct low-ACN NBR synthesis patent, it provides process analogues for acrylonitrile feed management in butadiene-based elastomer systems.

Patent 3

- Patent Number: US11813892B2
- Patent Title: Acrylated and acylated or acetalized polyol as a biobased substitute for hard, ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses synthesis of acrylated, acylated, or acetalized polyol monomers and their polymerization by controlled radical methods, including RAFT, to form homopolymers, copolymers, block copolymers, and statistical copolymers. The disclosed chemistry is not NBR-based, but it is a controlled polymerization platform for architecture and functionality tuning. - Glycerol: 1 mol — Example 2 - Methacrylic acid: 1.56 moles — Example 2 - Amberlyst: 5 g per 100 mL of solution — Example 2 - Reaction time: 24 hours — Example 2 - Reaction temperature: 100° C. — Example 2 - Glycerol: 1 mol, 10% mass by final solution mass of acetic acid — Example 3 - Acetic anhydride: 2 moles — Example 3 - Amberlyst: 5 g per 100 mL of solution — Example 3 - Stirring time: approximately 16 hours — Example 3 - Acetic anhydride addition time: 1 hour — Example 3 - Acetic anhydride: 3 moles per mole of 1-glycerol methacrylate — Example 4 - Room temperature: approximately 16 hours — Example 4 - Methacrylic anhydride: 2 moles per mole of 1,3 glycerol acetate — Example 5 - Reaction temperature: 80° C. — Example 5 - Reaction time: 4 hours — Example 5 - GRAMA: 3 g — Example 11 - Toluene: 15 g — Example 11 - CYCART: 0.006 g — Example 11 - AIBN: 0.000493 g — Example 11 - Argon purge: 15 minutes — Example 11 - Reaction temperature: 80° C. — Example 11 - Reaction time: 4 hours — Example 11 - SMA: 1 g — Example 14 - Toluene: 4 g — Example 14 - AIBN: 0.00115 g — Example 14 - CYCART: 0.00205 g — Example 14 - Reaction temperature: 80° C. — Example 14 - Reaction time: 4 hours — Example 14 - GCM: 1 g — Example 17 - Toluene: 2 g — Example 17 - AIBN: 0.00082 g — Example 17 - CYCART: 0.00205 g — Example 17 - Reaction temperature: 80° C. — Example 17 - Reaction time: 4 hours — Example 17 - GMM: 1 g — Example 20 - Toluene: 1 g — Example 20 - AIBN: 0.00143 g — Example 20 - CYCART: 0.00256 g — Example 20 - Reaction temperature: 80° C. — Example 20 - Reaction time: 4 hours — Example 20

Relevant Experimental Evidence - Example 2: Demonstrated esterification of glycerol with methacrylic acid at 100° C. for 24 h using Amberlyst. - Example 5: Showed preparation of a methacrylated/acylated glycerol monomer at 80° C. for 4 h. - Example 11: Demonstrated RAFT polymerization of GRAMA with CYCART and AIBN at 80° C. - Example 14: Showed free-radical polymerization of SMA at 80° C. for 4 h. - Example 17: Demonstrated polymerization of GCM under the same 80° C./4 h conditions.

Technical Relevance This patent is only indirectly relevant to low-acrylonitrile NBR, but it is useful as a controlled radical polymerization reference. It illustrates how monomer functionality and polymer architecture can be tuned through RAFT and related methods, concepts that are transferable to controlled NBR synthesis strategies.

Patent 4

- Patent Number: US9909003B2
- Patent Title: Fire-retardant composite materials
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses fire-retardant epoxy composite formulations that use carboxyl-terminated butadiene acrylonitrile rubber as a toughening additive in the resin matrix. The NBR-related component is used as a formulation ingredient rather than being synthesized in the patent. - Carboxyl-terminated butadiene acrylonitrile rubber: used as a toughening additive — Example a - Difunctional-terminated rubber: from 1 to 15 wt % of the weight of the resin matrix — Example a - Polyhydroxyl ether: from 2 to 15 wt % of the weight of the resin matrix — Example a - Rubber dispersion: from 3 to 15 wt % of the weight of the resin matrix — Example a - Ammonium polyphosphate: 15 to 30 wt % of the weight of the resin matrix — Example a - Zinc borate: 1 to 10 wt % of the weight of the resin matrix — Example a - Amorphous silicon dioxide: 0.1 to 1.5 wt % of the weight of the resin matrix — Example a - Matrix resin composition: 31 to 38 wt % non-halogenated bisphenol F epoxide resin — Example 1 - Phosphorus content in modified epoxide resin: 3 to 5 wt % added phosphorus — Example 1 - Tri-functional epoxide diluent: 5 to 15 wt % — Example 1 - Catalyst carrier: up to 10 wt % — Example 1 - Curing schedule: 100° C. for 1 hour, then 150° C. for 3 hours — Example 14 - Glass fiber content: 70 wt % — Example 14 - Viscosity at 21° C.: 34 sec — Example 14 - Delamination width: 5 mm — Example 14 - Viscosity of composition: 11 Pa·sec — Example 15 - Glass transition temperature: 150° C. — Example 15 - Thermal shock resistance: >50 cycles — Example 15 - Shear adhesion strength: 18 MPa — Example 16

Relevant Experimental Evidence - Example a: Identified CTBN-type butadiene acrylonitrile rubber as a toughening additive in fire-retardant epoxy systems. - Example 1: Showed that

phosphorus-modified epoxide plus mineral filler and imidazole curing agent improved FST performance. - Example 14: Demonstrated prepreg/composite fabrication at 100° C. and 150° C. with reduced delamination width. - Example 15: Showed a balance of low viscosity, Tg of 150° C., and thermal shock resistance >50 cycles. - Example 16: Demonstrated improved shear adhesion strength in toughened epoxy formulations.

Technical Relevance This patent is relevant because it explicitly uses butadiene acrylonitrile rubber as a toughening component, confirming the compatibility of NBR-derived elastomers in high-performance formulations. It does not teach NBR synthesis, but it is useful for understanding downstream application requirements for low-ACN NBR materials.

Patent 5

- Patent Number: US10947337B2
- Patent Title: Toughened epoxy resin composition
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses toughened epoxy resin compositions containing end-capped polyalkylene oxide and core-shell polymer, with comparative use of CTBN-type liquid tougheners in some formulations. The NBR-related content is formulation-based rather than synthetic, and the patent focuses on balancing viscosity, toughness, Tg, and adhesion. - End-capped polyalkylene oxide PTA3000: 5.0 parts by weight — Example 9 - End-capped polyalkylene oxide PMB3000: 5.0 parts by weight — Example 9 - End-capped polyalkylene oxide PMB1870: 5.0 parts by weight — Example 9 - End-capped polyalkylene oxide PMB1200: 5.0 parts by weight — Example 9 - End-capped polyalkylene oxide PPG2000: 5.0 parts by weight — Example 9 - End-capped polyalkylene oxide PGEB2000: 5.0 parts by weight — Example 9 - Epoxy resin BPADGE: 68.8 parts by weight — Example 9 - Epoxy curing agent IPD-D230: 19.5 parts by weight — Example 9 - Core shell polymer D-3P: 2.0 parts by weight — Example 9 - Total amount: 100.0 parts by weight — Example 9 - Viscosity at 25° C.: 29 to 30 sec — Example 9 - Fracture toughness GIC: 510 to 1150 J/m² — Example 9 - Glass transition temperature: 117 to 129° C. — Example 9 - End-capped polyalkylene oxide PTA3000: 6.0 parts by weight — Example 14 - Epoxy resin BPADGE: 35.3 to 56.0 parts by weight — Example 14 - Epoxy curing agent MTHPA-1% 2E4MZ: 38.7 to 44.0 parts by weight — Example 14 - Core shell polymer D-1: 6.0 to 12.0 parts by weight — Example 14 - Curing schedule: 100° C. for 1 hour, then 150° C. for 3 hours — Example 14 - Glass fiber content: 70 wt % — Example 14 - Delamination width: 5 to 12 mm — Example 14 - End-capped polyalkylene oxide A-3: 6.0 or 8.0 parts by weight — Example 15 - Epoxy resin BPADGE: 18.8 to 28.0 parts by weight — Example 15 - Epoxy curing agent C-1: 18.5 to 22.0 parts by weight — Example 15 - Core shell polymer D-1: 0.0 to 8.0 parts by weight — Example 15 - Filler CMC-12S: 49.9 to 50.0 parts by weight —

Example 15 - Defoaming agent BYK-1790: 0.2 parts by weight — Example 15 - Viscosity of composition: 10 to 33 Pa·sec — Example 15 - Glass transition temperature: 150° C. — Example 15 - Thermal shock resistance: 10 to >50 cycles — Example 15 - End-capped polyalkylene oxide A-1: 5.0 or 10.0 parts by weight — Example 16 - Epoxy resin BPADGE: 47.8 to 66.1 parts by weight — Example 16 - Epoxy curing agent C-3: 30.5 to 33.9 parts by weight — Example 16 - Core shell polymer D-1: 0.0 to 9.7 parts by weight — Example 16 - Liquid toughener CTBN-Adduct: 12.5 or 25.0 parts by weight — Example 16 - Liquid toughener DY965: 5.0 or 10.0 parts by weight — Example 16 - Thixotropic agent TS-720: 2.5 parts by weight — Example 16 - Silane coupling agent Z-6040: 0.5 parts by weight — Example 16 - Viscosity of composition: 9 to 18 MPa? — Example 16 - Shear adhesion strength: 9 to 18 MPa — Example 16

Relevant Experimental Evidence - Example 9: Showed that different end-capped polyalkylene oxides combined with BPADGE and IPD-D230 gave low-viscosity toughened epoxy compositions. - Example 14: Demonstrated prepreg fabrication and low delamination width after curing at 100° C. and 150° C. - Example 15: Showed that the composition could maintain Tg at 150° C. while varying core-shell polymer content. - Example 16: Demonstrated adhesion performance in formulations containing CTBN-type liquid tougheners and core-shell polymer. - Example 9: Reported fracture toughness values spanning 510 to 1150 J/m² across formulations.

Technical Relevance This patent is only indirectly relevant to low-acrylonitrile NBR because it uses CTBN-type butadiene acrylonitrile liquid rubber as a comparative or auxiliary toughener. It is useful for understanding how NBR-derived elastomers are deployed in epoxy systems and what property balance is expected from such materials.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US9023914B2 is the only patent in the set that directly discloses NBR synthesis; it uses solution free-radical copolymerization of acrylonitrile and butadiene in organic solvent, whereas the other patents are mainly formulation or graft-copolymer documents.
- The acrylonitrile feed in US9023914B2 is explicitly 37 phm in Example 1, with butadiene at 63 phm; this is a relatively butadiene-rich feed, but still not clearly in the low-ACN range by the target definition.
- US9023914B2 uses dimethylacetamide as solvent, Vazo® 88 initiator, DoPAT modifier, 100° C. polymerization temperature, 3 bar butadiene pressure, and 9 h reaction time; these are the most actionable synthesis conditions in the set.
- US9624333B2 discloses aqueous emulsion grafting of styrene/acrylonitrile onto agglomerated butadiene rubber, with potassium persulfate initiation and 68° C. to 80° C. processing; it is process-adjacent rather than a direct NBR recipe.
- US9909003B2 and US10947337B2 both use butadiene acrylonitrile rubber or CTBN-type tougheners in epoxy formulations, showing downstream compatibility of NBR-derived elastomers but not monomer-feed control for low-ACN synthesis.

- US11813892B2 is mechanistically unrelated to NBR but provides controlled radical polymerization concepts (RAFT, chain-transfer control) that are relevant as broader synthesis tools for architecture control.
- Across the set, the recurring process themes are monomer ratio control, temperature control, and molecular-weight management; however, only US9023914B2 explicitly couples these to NBR preparation.
- Coagulation/precipitation is explicitly disclosed in US9023914B2 (ethanol precipitation) and US9624333B2 (magnesium sulfate precipitation), while the epoxy patents focus on curing and composite fabrication rather than polymer isolation.

4. CONCLUSION

The patent evidence indicates that the most credible route to low-acrylonitrile NBR is controlled free-radical copolymerization of butadiene and acrylonitrile, preferably in solution when precise molecular-weight and architecture control are needed. US9023914B2 provides the strongest direct guidance: acrylonitrile/butadiene feed control, 100° C. polymerization, inerting, modifier-assisted chain control, and precipitation isolation. For low-ACN targeting, the evidence supports reducing acrylonitrile feed below the 37 phm example level while maintaining the same process discipline around temperature, pressure, and modifier loading. The remaining patents are valuable mainly as application and process analogues: US9624333B2 for emulsion grafting and acrylonitrile incorporation in rubber-rich systems, and US9909003B2/US10947337B2 for downstream use of NBR-derived tougheners in epoxy composites. Recommended synthesis parameters based on the evidence are solution polymerization in an oxygen-free organic solvent, moderate-to-high temperature around 100° C., controlled butadiene feed under pressure, and explicit monomer-ratio adjustment to lower acrylonitrile content while monitoring conversion and molecular weight.

5. REFERENCES

- US9023914B2 | Nitrile rubbers and production thereof in organic solvents | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9023914B2/en>
- US9624333B2 | Thermoplastic moulding materials based on acrylonitrile, styrene and butadiene | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9624333B2/en>
- US11813892B2 | Acrylated and acylated or acetalized polyol as a biobased substitute for hard, ... | Not disclosed | US | 2023 | <https://patents.google.com/patent/US11813892B2/en>
- US9909003B2 | Fire-retardant composite materials | Not disclosed | US | 2018 | <https://patents.google.com/patent/US9909003B2/en>
- US10947337B2 | Toughened epoxy resin composition | Not disclosed | US | 2021 | <https://patents.google.com/patent/US10947337B2/en>